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Co-firing hydrogen with natural gas for 9 MW Siemens Energy dry low emissions combustor: Part I – high pressure rig test

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ABSTRACT

To validate 30% H_2 (by volume) capability for the Siemens Energy's 9 MW SGT-300-2S engine, high-pressure rig testing was conducted. The dry low emissions combustor was initially operated with natural gas to establish a baseline operation. Subsequently, it was fuelled with a 30% H_2 (by volume) blend under various ambient temperatures. To evaluate operational margins, tests were extended to 35% and 40% H_2. Finally, the combustion system's robustness was rigorously tested under various of overfiring conditions. The impact of H_2 concentration on combustor performance was investigated, focusing on factors such as burner metal temperature, NO_x and CO emissions, combustion dynamics, and combustor wall temperatures. Test results indicate that burner metal temperatures and NO_x emissions increase with increasing H_2 concentration. The turbine entry temperature at which CO emissions begin to rise is lowered with increasing H_2 concentration. This suggests that adding H_2 to the fuel can lower the turndown temperature for controlling CO emissions, thereby improving the engine's part load thermal efficiency. Combustion dynamic pressure amplitudes remained low for H_2 concentrations up to 40%. Combustor wall temperatures for H_2 operation up to 40% were comparable to those for natural gas. Test results demonstrate that the combustion system can safely operate with up to 40% H_2 by volume in the fuel. The effect of pilot split on combustion was investigated for natural gas and the fuel blends with 30% and 40% H_2. A key finding is that different fuel compositions may require different pilot split and turndown control temperature for optimal operation.

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1. INTRODUCTION

Gas turbines (GTs) are vital in fulfilling the world's energy needs, serving both power generation and industrial applications. However, their operation with fossil fuels releases carbon dioxide (CO_2) emissions, contributing to climate change. Utilising hydrogen (H_2) as a fuel offers a potential solution to reduce CO_2 emissions in gas turbine applications. Research and development efforts are actively pursuing this goal [1-3]. While transitioning to 100% H_2 faces challenges related to economics, materials, storage, and combustion process [4-9], blending H_2 with natural gas (NG) presents a viable pathway to gradually increase H_2 usage in gas turbines [9]. This approach allows for continued progress towards a cleaner energy future while addressing the current limitations of H_2 production and utilisation.

H_2 is a highly reactive gas that can significantly alter the combustion properties of fuels. When added to fuels, H_2 can increase reactivity, broaden flammability limits, and raise adiabatic flame temperature and laminar flame speed [7, 9]. Consequently, burning fuels containing H_2 generally results in higher NO_x emissions and lower CO emissions [10-12]. These experimental investigations were conducted across a variety of test setups. They included a scaled-down industrial gas turbine Dry Low Emissions (DEL) combustor operated within a higher-pressure test rig [10], an industrial concept DLE combustor tested in an atmospheric pressure rig [11], and different combustor designs developed specifically for engine integration [12]. This diverse range of test environments provides valuable insights into the combustion behaviour of H_2-enriched fuels across different operating conditions and combustor configurations. For example, observations from [10] suggest that NO_x emissions may increase with increasing H_2 content in the fuel. In some cases, NO_x emissions were observed to increase by 25% with 10% H_2 addition and by 50% with 70% H_2 addition. Conversely, CO emissions were observed to decrease by 50% with 90% H_2 addition. This shift in emissions is primarily due to higher flame temperatures and changes in chemical reaction pathways.

Recent research by Cole *et al*. [13] highlights the importance of considering CO emissions when reporting the emissions of non-carbon-containing fuels such as pure H_2. This emphasis arises from the recognition that even in the combustion of pure H_2, trace amounts of CO_2 present in the ambient air can

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react to form CO under fuel-rich combustion conditions. This phenomenon underscores the need for a comprehensive understanding of all relevant emissions, even for nominally carbon-free fuels.

Furthermore, fuels containing H_2 have the potential to increase the combustion dynamic pressure amplitude, especially at high H_2 concentrations [14]. This can lead to higher frequencies and amplitudes of pressure oscillations [12], potentially destabilising the combustion process. Additionally, H_2 enriched fuels are more susceptible to flashback, a phenomenon where the flame propagates backward into the burner [15, 16]. The risk of flashback increases with higher H_2 concentrations, as the increased reactivity can cause the flame to attach to the burner.

Figure 1 illustrates the changes in fuel properties, in terms of Lower Heating Value (LHV) and Wobbe Index (WI), due to H_2 blending with natural gas. Figure 2 presents the resulting adiabatic flame temperature and laminar flame speed under typical gas turbine engine operating conditions for various volumetric H_2 blending ratios.

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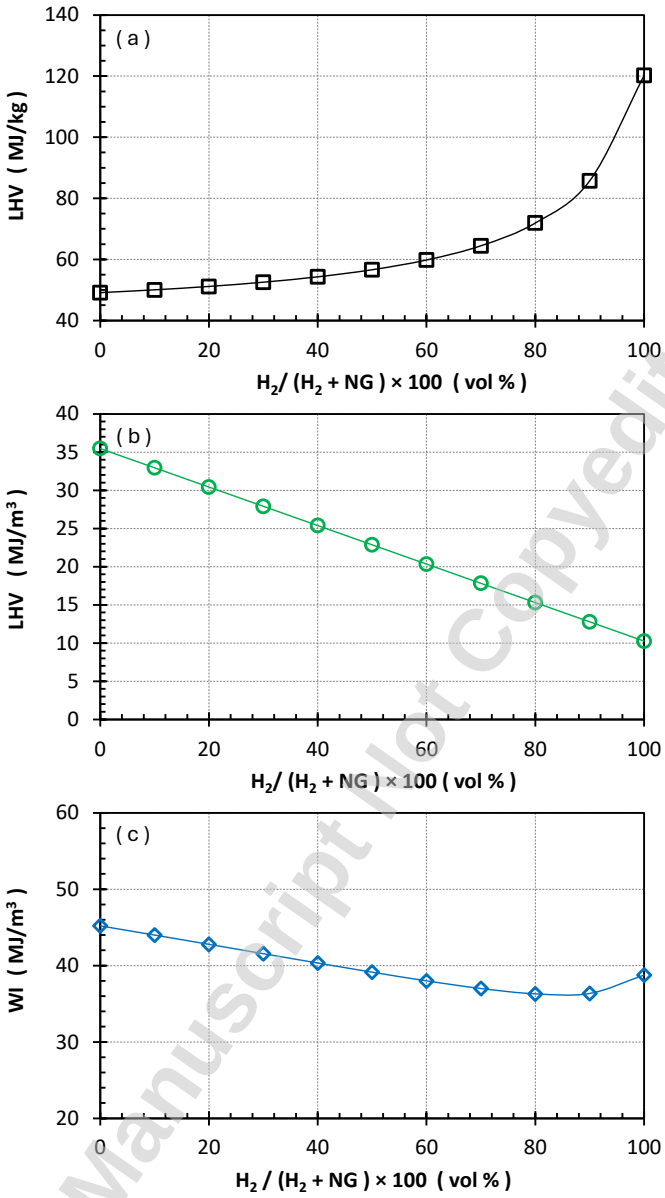

Figure 1 LHV (top figure: mass-based; middle figure: volume-based) and WI as a function of H_2 volumetric concentration for natural gas and H_2 blends at ISO conditions

Figure 1 illustrates the variation of LHV for natural gas and H_2 blends at ISO (standard) conditions (1.0135 bar and 15 °C), on a mass basis (Fig. 1(a)) and volumetric basis (Fig. 1(b)) as a function of H_2 volumetric concentration in the fuel blend, where 0 indicates pure natural gas (without any H_2 addition) and 100 indicates pure H_2. The WI and mass-based LHV were obtained from Liu *et al*. [17]. The volume-based LHV was calculated by multiplying the mass-based LHV by the fuel blend density. It is shown that the mass-based LHV of the fuel blend exhibits a near-linear increase with H_2 concentration

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when H_2 concentration is lower than 60%. Once the H_2 concentration exceeds 60%, it increases exponentially with H_2 concentration. Conversely, the volume-based LHV decreases linearly with H_2 concentration. The observed trends suggest that for a gas turbine engine operating at constant heat input, reducing the fuel mass flow rate becomes necessary as the H_2 concentration in the fuel blend increases. However, the fuel volume flow rate increases with increasing H_2 concentration to maintain the same heating requirement. Notably, when operating with 100% H_2, a substantial decrease in fuel mass flow rate and a substantial increase in volume flow rate can be expected relative to pure natural gas. Figure 1(c) illustrates the variation of WI within the fuel blend. As H_2 is added, the WI initially decreases. However, once the H_2 concentration surpasses 80%, the WI begins to increase with further H_2 additions. These features suggest that handling and mixing of H_2 with natural gas may pose significant challenges for GT operators.

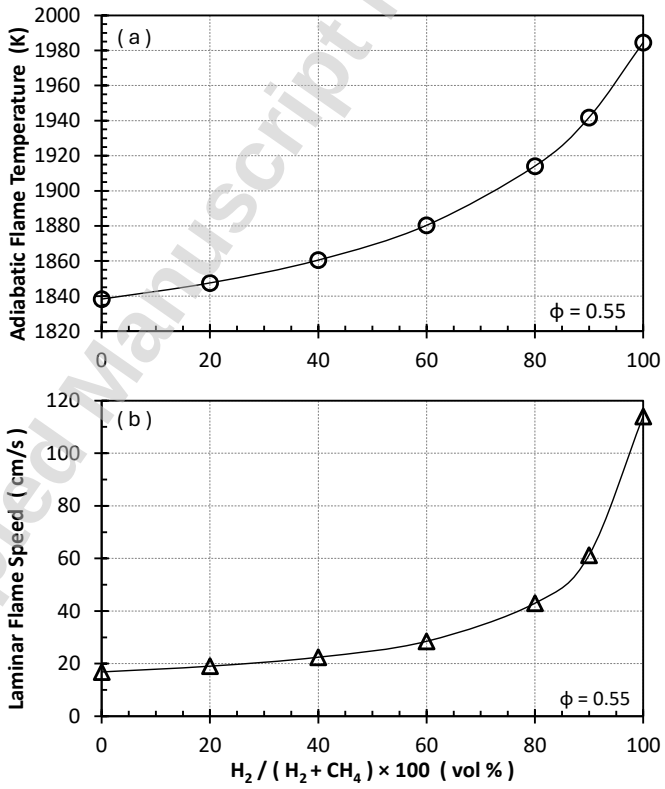

Figure 2 Adiabatic flame temperature and laminar flame speed at generic gas turbine combustor operating conditions: 14 bar, 666 K and equivalence ratio of 0.55

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Adiabatic flame temperature and laminar flame speed calculations were conducted for methane (CH_4) and H_2 blends operating under conditions representative of a generic 9 MW gas turbine combustor [17]. Given the comparable adiabatic flame temperature and laminar flame speed of CH_4 and natural gas under these conditions [17], Figure 2 can also be used to represent the corresponding values for natural gas and H_2 blends. The fuel-air mixture was maintained at a pressure of 14 bar, a temperature of 666 K, and an equivalence ratio of 0.55. Adiabatic flame temperatures were calculated using NASA equilibrium code [18], while laminar flame speeds were computed using Ansys Chemkin Pro (Version 2022 R1) and the GRI 3.0 chemical kinetics mechanism [19]. The results of these calculations are depicted in Figure 2.

As H_2 concentration increases in the fuel blend, both flame temperature and laminar flame speed increase accordingly. While the increase in laminar flame speed with H_2 concentration below 60% remains relatively steady, an exponential increase is evident as H_2 concentration surpasses 60%. Notably, when operating with 100% H_2, a substantial enhancement in laminar flame speed is observed relative to pure CH_4. The elevated flame temperature and increased laminar flame speed associated with fuel blends containing higher concentrations of H_2 present significant challenges for gas turbine combustor design.

Siemens Energy's 5 MW to 15 MW industrial gas turbine engines are now capable of operating on a H_2-enriched natural gas fuel containing at least 30% H_2 by volume. To prove the 30% H_2 capability of the 9 MW SGT-300-2S engine, high-pressure rig testing was conducted at the Siemens Energy Clean Energy Center (CEC) in Berlin.

After the baseline operation with natural gas, the combustor was fuelled with a 30% H_2 blend fuel under various ambient temperature conditions, including ISO, -20 °C, and +45 °C. The focus was on factors such as burner metal temperature, NO_x and CO emissions, combustion dynamics, and combustor wall metal temperature.

To assess operational margins, tests were extended to 35% H_2 and 40% H_2 under ISO conditions. To test the robustness of the combustor, it was also tested under various overfiring conditions to establish transient operability with H_2 in the fuel.

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This Part I paper presents the high-pressure rig test results of the SGT-300-2S combustion system with H_2 volume concentrations up to 40%. Numerical simulations to gain deeper insights into the underlying physics are reported in Part II [20].

2. CURRENT STUDY

The DLE combustion system of a 9 MW Siemens Energy SGT-300 engine was extensively tested in a high-pressure combustion rig at engine operating conditions. This part I paper reports the high-pressure rig test results on various fuel blends under various operating conditions. Numerical simulations were carried out in part II for selected fuel blends: natural gas, and with 30% and 40% H_2 at ISO full load condition. Impact of pilot split was also studied for the fuel blend with 40% H_2. Large Eddy Simulations with a Flamelet Generated Manifold combustion model and Thickened Flame Model were employed. The ARM22 chemical mechanism was used to generate flamelet manifolds for each configuration of varied H_2 concentration.

This section provides a detailed overview of the test facility, test hardware, and test conditions employed in the experiment.

2.1 DLE Combustion System

The test vehicle for this research was the DLE combustion system of the SGT-300 twin-shaft engine, a member of the versatile SGT-300 engine family. Notable for its robustness, reliability and adaptability, the SGT-300 engine family can be employed in various applications, including mechanical drive, power generation, and combined heat and power. The family offers two distinct configurations:

- **Single-shaft engine (SGT-300-1S):** Generating a power output of 7.9 MWe.
- **Twin-shaft engine (SGT-300-2S):** Delivering a power output of up to 9.0 MW for mechanical drive and 8.5 MWe for power generation.

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As a standard, the engine family operates on gaseous fuels with a WI ranging from 25 MJ/m³ to 65 MJ/m³ (not including H_2 or H_2 enriched fuels). It is well-regarded for its stringent emissions guarantees, achieving NO_x and CO emissions guarantees of 15 and 10 parts per million respectively, corrected to 15% O_2 concentration, on a volumetric and dry basis (ppmvd), with the option of sub 10 ppmvd NO_x [21].

The SGT-300-2S engine demonstrates a notable capability for part load operation by adjusting fuel and air flow rates without requiring variable guide vane (VGV) or bleed-to-exhaust (B2E) modulation. This characteristic is referred to as "natural turndown" in this paper. Additionally, the engine can be operated at part load through B2E modulation, a technique that bypasses compressor exit air into the exhaust to maintain optimal CO emissions. This method is termed "turndown control" in this paper.

Designed as a reverse-flow can-annular combustion system, the SGT-300 combustor is based on the proven DLE combustor technology for gaseous and liquid fuel operation. As illustrated in Figure 3, it consists of three main sections:

(1) **Pilot burner:** Houses the pilot fuel galleries and injectors for both gaseous and liquid fuel.

(2) **Main burner:** Houses the main air swirler and main gaseous and liquid fuel system.

(3) **Combustor can:** Includes the premixing chamber (prechamber) and a double-skin impingement-cooled liner.

Fuel is introduced in two stages, as shown in Figure 3. The main fuel is injected into the swirler air path for optimal premixing and reduced NO_x emissions. Pilot fuel is injected at the end of the swirler air path to create a diffusion flame and stabilise the combustion process. The pilot split, or pilot schedule, is the ratio of pilot fuel usage to total fuel usage.

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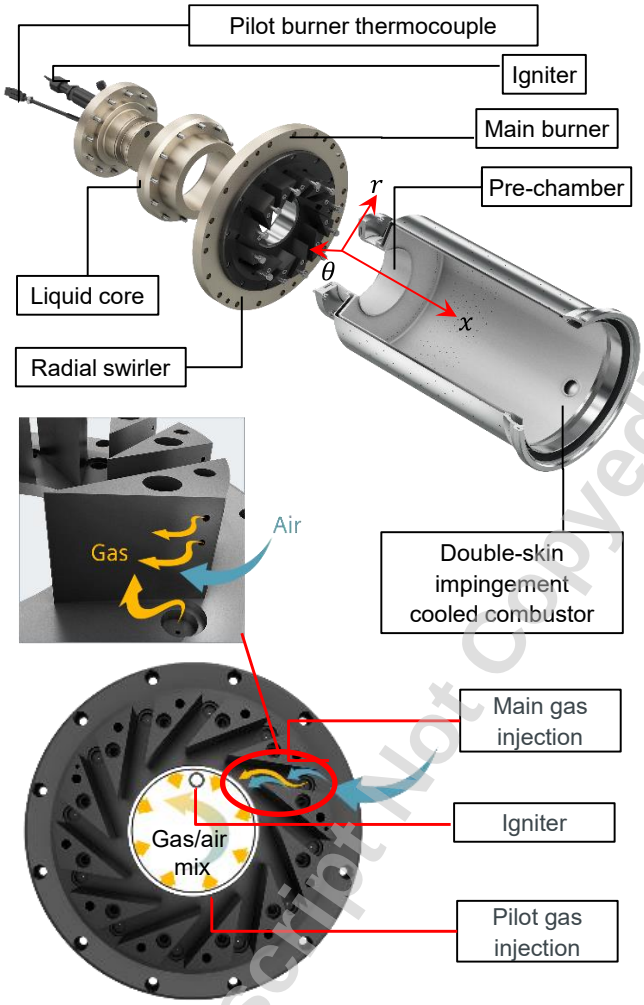

Figure 3 Siemens Energy DLE combustor configuration, air paths and fuel injection

Downstream of the combustor can, a transition duct is used to adapt the flow from the circular combustor geometry to a sector of the turbine entry annulus.

The main aerodynamic features of the combustor, such as fuel placement, and heat release location are highlighted in Figure 4. The main combustion air enters through a single radial swirler at the head of the combustor, then turns through a right angle into the prechamber, followed by a sudden expansion into the combustion chamber. The swirl number is sufficiently high to induce a vortex breakdown reverse flow zone along the axis, forming the central recirculation zone (CRZ). This CRZ remains attached to the pilot burner face, establishing a firm aerodynamic base for flame stabilisation. In the wake of the sudden expansion, an outer recirculation zone (ORZ) is established. The flame is stabilised in the shear layers formed by the CRZ and ORZ.

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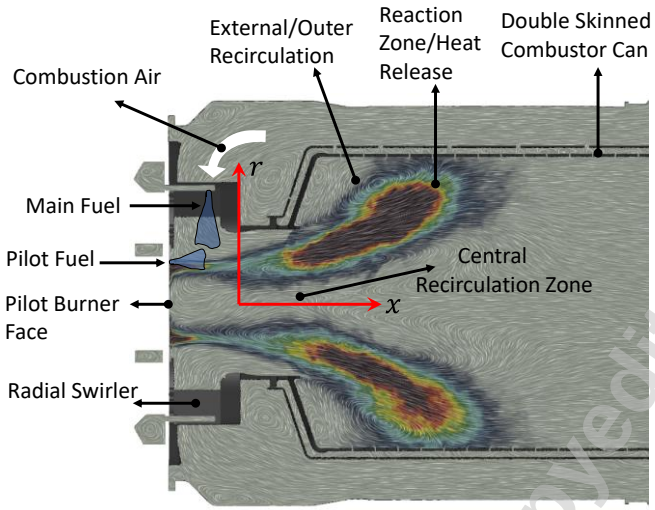

Figure 4 Cross-section of the DLE combustor showing key aerodynamic features

Recently the pilot burner has been optimised through optimising the pilot gaseous fuel injection holes design, resulting in significant reduction in the effective flow area leading to significantly lower pilot fuel usage while maintaining stable fuel flow, especially at part load. The benefits of the optimised pilot burner include reduction in pilot fuel usage, enhanced fuel-air mixing, and stabilised fuel flow at part load, as a result, NO_x emission is significantly reduced, and combustion dynamic pressure at part load is improved [21].

The DLE combustion system studied in the paper include standard main burner, optimised pilot burner, standard combustor can and transition duct.

2.2 High-Pressure Combustion Rig

The Siemens Energy high-pressure combustion rig, located at the CEC in Berlin, Germany, as detailed in [8, 22], is a single combustor test facility designed to replicate actual engine operating conditions for combustor development purposes.

The CEC houses three specialised test cells designed for rigorous combustion system evaluations. This versatile setup accommodates a wide range of testing needs, from large gas turbine burners to smaller industrial counterparts. The CEC can accurately replicate actual engine operating conditions by:

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- **High-capacity infrastructure:** Utilising three compressors and two advanced air preheating systems to support demanding tests. This setup enables precise adjustments of air pressure (up to 40 bar) and temperature (up to 600°C), with an air flow rate of up to 50 kg/s.
- **Fuel flexibility:** Adapting to various fuels, including liquid fuels such as diesel, kerosene, and methanol, gaseous fuels such as natural gas, propane, and H_2. It also allows for blending different fuels to meet specific requirements. This blended fuel is then seamlessly integrated into fuel streams of burner.
- **Monitoring performance:** Continuously measuring and analysing critical factors such as emissions, flame stability, and component metal temperatures, including optical visual flame observation to detect flash back particularly important for fuel blend with H_2.
- **Advanced instrumentation:** Employing state-of-the-art measurement devices and a sophisticated data acquisition system for accurate and reliable results.
- **Precise control:** Ensuring high accuracy in critical parameters, such as hydrogen mass flow rate, with tight control tolerances.

This comprehensive testing environment enables the CEC to provide valuable insights and support the development of efficient and environmentally friendly combustion technologies.

The test Cell 3 was used for current study. The integration of a combustor into the test rig is illustrated in [8]. For the Cell 3, the H_2 mass flow accuracy can be controlled within +/- 0.25% when H_2 flow rate exceeds 2g/s.

The combustion hardware of SGT-300-2S was extensively instrumented to measure various operating parameters including fuel supply pressures and temperatures, air pressure and temperature, hardware metal temperatures, emissions, and combustion dynamics. A thermocouple was fitted at the center of the pilot burner to monitor any changes to the pilot burner surface temperatures. Moreover, a multitude of thermocouples were strategically mounted at critical locations on the combustor can to monitor and record the temperature pattern and identify potential flashback risk. Emissions measurements included NO_x, CO, and Unburnt Hydrocarbons (UHC) and the measured emissions were

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corrected to 15% O_2 on a volumetric and dry basis. Combustion dynamics signature was measured by a dynamic pressure sensor. All measurements were logged in the data acquisition system for subsequent analysis.

2.3 Test Conditions

The combustor's performance was initially evaluated at ISO ambient conditions using natural gas as a baseline. Subsequently, blended fuels containing natural gas and varying H_2 concentrations of 30%, 35%, and 40% by volume were introduced to assess the combustor's operability with different H_2 blends.

To evaluate the impact of H_2 on turndown control temperature, engine operating conditions were systematically varied from full power to part load under natural turndown conditions. Following this, turndown control temperature was optimised for each H_2 concentration. Additionally, the influence of pilot split on combustion performance was investigated using natural gas, 30% H_2, and 40% H_2 blends. The effect of ambient temperature on combustor performance was assessed with a 30% H_2 fuel mixture. Finally, the combustor was tested under various overfiring conditions to assess its robustness and establish transient operability with hydrogen in the fuel.

3. RESULTS AND DISCUSSIONS

3.1 Test Results under Natural Turndown Operation

This section outlines the rig test results conducted at ISO ambient conditions under natural turndown operation.

Pilot splits across all engine load conditions tested, as a function of turbine entry temperature (TET), are shown in Figure 5 for each fuel blend. To minimise the impact of pilot splits on the overall test results, their values were kept constant at the same TET. Pilot splits and TET were normalised against their full-load values. At higher TET, significantly lower pilot splits were used to reduce NO_x emissions. To enhance flame stability and reduce CO and UHC emissions, the pilot split was typically increased at lower TET.

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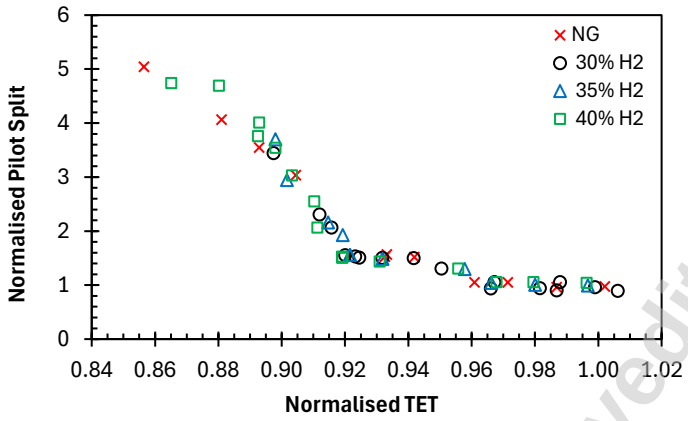

Figure 5 Pilot splits across engine loads for all fuel blends tested

NO_x and CO emissions plotted against TET are shown in Figure 6 for all the fuel blends investigated. TET was normalised by its full load value. The results indicate that at the same TET and similar pilot splits, NO_x emissions are increased for fuel blends containing H_2 as a result of higher flame temperatures. Among the tested blends, 40% H_2 produced the highest NO_x emissions across all engine load conditions. No significant difference was observed between 30% and 35% H_2 blends. Less than 25 ppmvd NO_x emissions were achieved for natural gas at all engine loads tested. NO_x emissions below 25 ppmvd could be achieved with H_2 blends from full load down to TET 0.92, except for 40% H_2 at full load. However, lowering the pilot split could potentially reduce NO_x emissions in this case as demonstrated later in the paper.

As shown in Figure 6(b), CO emissions were negligible at higher TETs. The 'kick-off' point, i.e., where CO emissions began to rise exponentially, was delayed for H_2 blends as the result of higher flame temperatures. Among the H_2 blends, 35% and 40% exhibited similar kick-off points, which were lower than that of the 30% H_2 blend. In this paper, symbols represent the test results, while lines indicate the overall trends.

To ensure NO_x and CO emissions remained below 25 ppmvd and 10 ppmvd, respectively, the turndown control temperature, expressed in terms of normalised TET, was determined to be approximately 0.94 for natural gas based on CO emissions and around 0.92 for H_2 blends based on NO_x

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emissions. Lowering the turndown control temperature can effectively enhance engine thermal efficiency at part load. The turndown control temperature for 40% H_2 blend could be further optimised by reducing the pilot split to reduce NO_x emissions, especially at TET values below 0.92 where CO emissions were lower than 10 ppmvd but NO_x emissions exceeded 25 ppmvd. The optimised control temperature will be discussed in detail later in the paper.

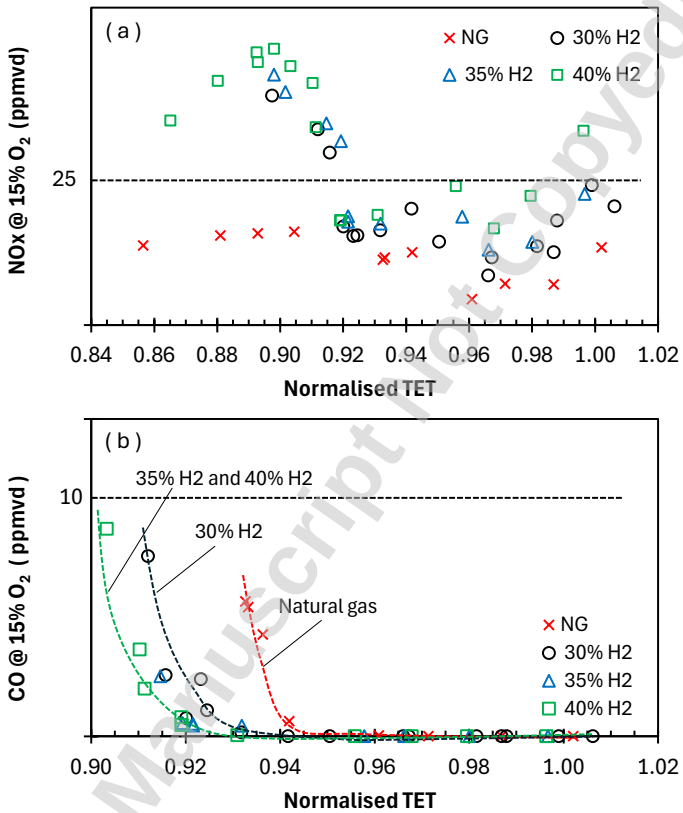

Figure 6 Impact of H_2 concentration on NO_x and CO emissions

As previously mentioned, a thermocouple was installed at the center of the pilot burner to monitor the pilot burner metal temperature on the surface exposed to the hot combustion side. The recorded temperatures for different engine loads with natural gas and various H_2 blends are presented in Figure 7. Both metal temperatures and TETs were normalised against natural gas full load values. For all fuels tested, pilot burner surface temperature decreased as engine load decreased and were within design limit for safe operation. Compared to natural gas, a significant increase in pilot burner surface temperature was observed for H_2 blends when TET exceeded 0.92. Notably, higher pilot burner surface temperatures

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were observed with increasing H_2 concentration. However, below TET 0.9, the temperature increase for H_2 blends was relatively small.

The increased surface temperature of the pilot burner can be attributed to two primary factors. Firstly, fuels containing H_2 exhibit higher flame temperatures, which directly impact the heat transfer to the burner surface. Secondly, the higher reactivity and increased flame speed of H_2 containing fuels results in more compact flames that propagate upstream [14], bringing them closer to the pilot burner. This proximity intensifies the heat transfer to the burner surface, further contributing to the observed temperature increase.

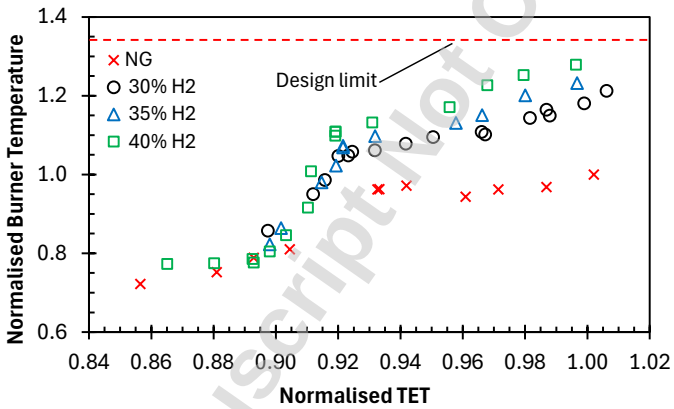

Figure 7 Effect of H_2 concentration on pilot burner surface metal temperature

Root mean square (RMS) values of the combustion dynamic pressure amplitudes, plotted against engine load, are presented in Figure 8(a) for the frequency range of 0-100 Hz and Figure 8(b) for the frequency range of 600-1000 Hz. The low frequency dynamics (LFD) range of 0-100 Hz is associated with Lean Blow Out (LBO), where excessively high amplitudes can lead to flame extinction. While, the intermediate frequency dynamics (IFD) range of 600-1000 Hz represents high-energy pulsations due to coupled interaction between unsteady heat release and pressure fluctuations [14], which, if excessive, can potentially result in hardware damage. Pressure amplitudes were normalised against their design limits for both frequency ranges. A normalised value of 1 represents the design limit. As before, the TET reference point was the full load value.

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Combustion dynamic pressure amplitudes were noticeable higher for H_2 blends compared to natural gas, particularly at higher engine loads, such as when TET exceeded 0.92. The increase in combustion dynamic pressure amplitude was directly correlated with the concentration of H_2 in the blend, with higher H_2 concentrations leading to higher pressure amplitudes. However, it's important to note that even with these elevated amplitudes, they remained within the design limits, ensuring operational safety. It should be noted that no dominant frequency change was observed in either frequency range for fuel blends with H_2 compared to natural gas operation.

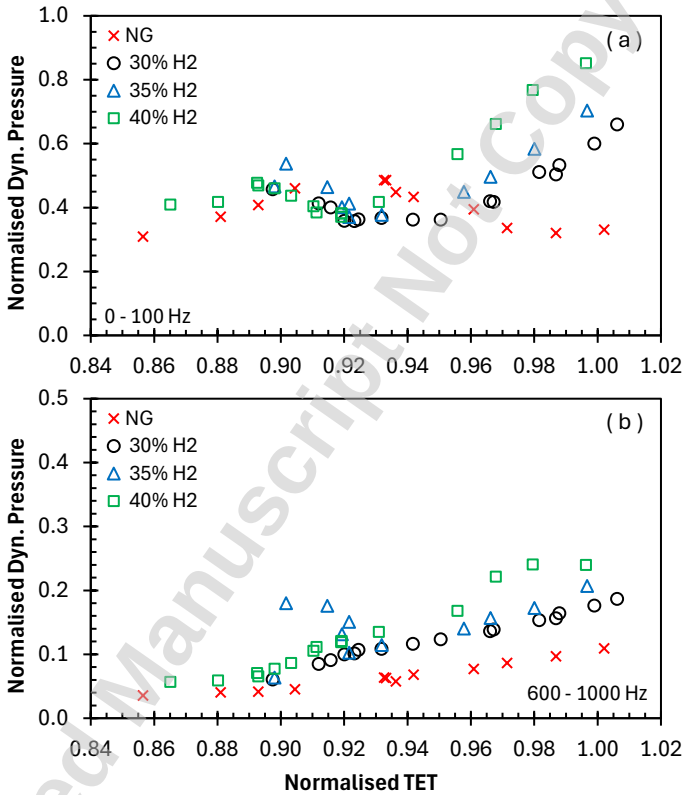

Figure 8 Influence of H_2 concentration on combustion dynamic pressure amplitude (top: LFD [0 – 100 Hz], bottom: IFD [600 – 1000 Hz])

A total of 44 thermocouples were installed on the combustor can, distributed across 11 rows along the combustor's axial coordinate as shown in Figure 9. Rows 1 and 3 had three thermocouples each, while rows 2 and 6 - 11 had four thermocouples each. Row 5 had only two thermocouples. Row 4 had a higher number of thermocouples with eight sensors. Within each row, the thermocouples were evenly spaced around the circumference.

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The wall temperatures measured by the thermocouples in each row are presented in Figure 9, with the values averaged across the thermocouples and normalised by the value at row 8 where the highest heat release from the flame occurs [23]. Here, X represents the total length of the combustor, and x is its axial coordinate. The results indicate that the wall temperatures in rows 1-3 and 7-11 were not significantly influenced by the H_2 concentration in the fuel. However, in row 4, wall temperature sensitivity to H_2 concentration was observed only above 30%, with 40% H_2 leading to higher wall temperatures. Interestingly, in rows 5 and 6, natural gas exhibited higher wall temperatures compared to H_2 blends, while no significant difference in wall temperature was found among various hydrogen blends. This phenomenon is explained by the numerical findings in Part II, which indicate that the addition of H_2 results in a two-fold change in the flame: an upstream shift and an increased distance from the wall as the flame angle decreases [20].

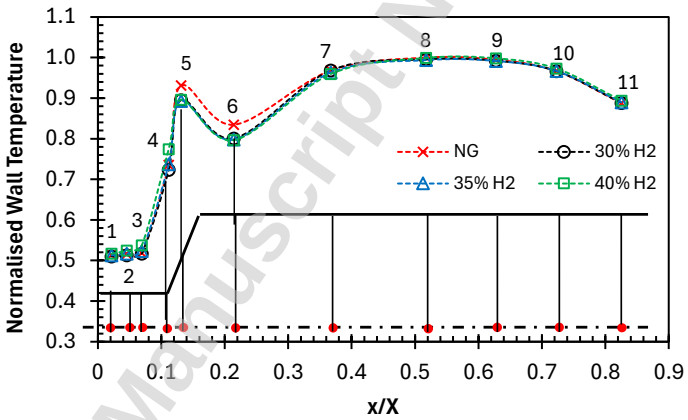

Figure 9 Impact of H_2 concentration on combustor can wall metal temperature: X signifies the overall length of the combustor, and x is the coordinate along its axis

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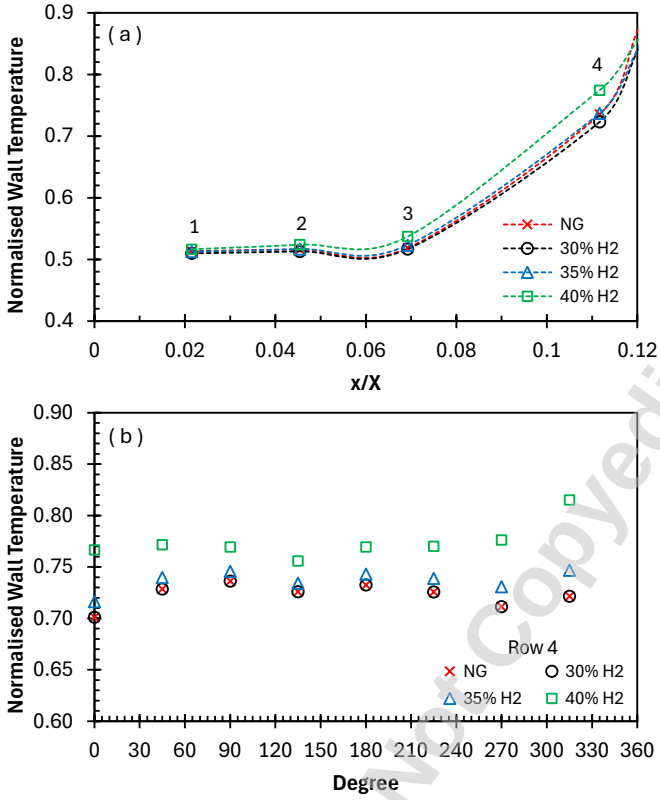

Figure 10 Impact of H_2 concentration on combustor prechamber wall metal temperature: X signifies the overall length of the combustor, and x is the coordinate along its axis

Figure 10(a) presents a magnified view of Figure 9, focusing on the prechamber wall temperatures at rows 1 - 4. Notably, there were no significant differences in wall temperature between natural gas and H_2 blends containing up to 30% H_2. However, with 40% H_2, wall temperatures increased, although this effect was less pronounced in rows 1-3 compared to row 4.

Figure 10(b) illustrates the wall temperature profile at row 4 which is the end of the prechamber. The profile was relatively uniform, and no significant differences were observed between natural gas and H_2 blends up to 30%. However, with 40% H_2, wall temperatures were higher compared to natural gas and lower H_2 blends, and the temperature profile exhibited greater fluctuations. This indicates the expected upstream shift in flame position due to higher flame speed with higher H_2 addition.

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3.2 Impact of Pilot Split on Combustion Performance

The impact of pilot split on combustor performance was investigated for natural gas and H_2 blends with 30% and 40% H_2 at full load conditions. Figure 11 illustrates the effect of pilot split on NO_x emissions and pilot burner surface temperature. Pilot splits were normalised relative to their default full load value, while NO_x emissions and pilot burner surface temperatures were normalised based on their natural gas full load values.

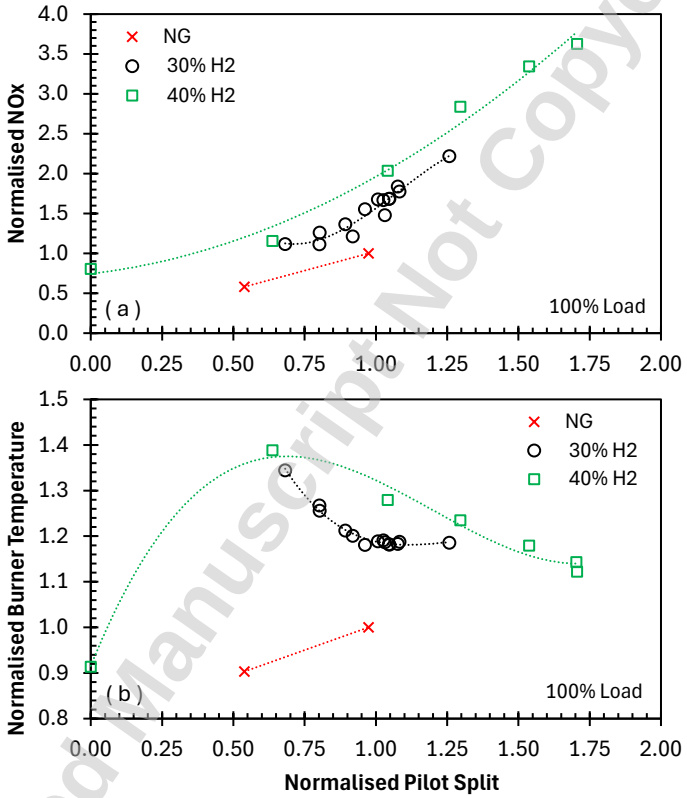

Figure 11 Effect of pilot split on NO_x emissions and pilot burner surface temperature

For natural gas, as expected, both NO_x emission and pilot burner surface temperature decreased as the pilot split was reduced. Similarly, for H_2 blends, NO_x emission decreased with decreasing pilot split. However, with H_2 blends, reducing the pilot split from its default value led to an increase in pilot burner surface temperature. Conversely, increasing the pilot split from its default value could effectively reduce pilot burner surface temperature, as shown for 40% H_2. For 30% H_2, pilot surface temperature remained relatively constant from the default pilot split up to 1.25 times its default value. Additionally, a zero

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percent pilot split was explored for 40% H_2, resulting in a significant reduction in both NO_x emission and pilot surface temperature. The reduction in NO_x emissions as the pilot split decreases is due to a reduction in the diffusion flame.

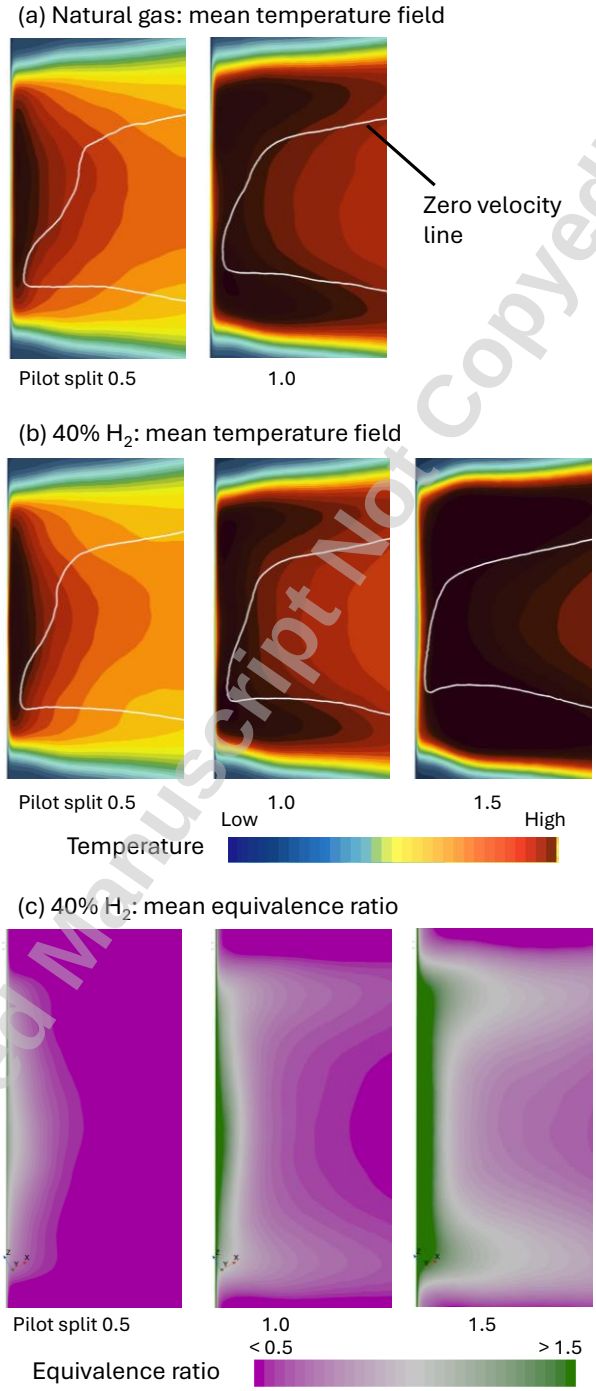

Figure 12 Cross-section contour plots in the vicinity of the pilot burner face for different pilot splits: mean temperature field with zero velocity line, for (a) natural gas, (b) 40% H_2, and (c) mean equivalence ratio for 40% H_2 [20]

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An analysis of flow field and equivalence ratio near the burner face (Part II [20], Figure 16) explains the reduced pilot burner temperature with increased pilot split for a 40% H_2 fuel blend. These results are cited here and shown in Figure 12. Figure 12 shows cross-sectional contour plots for different pilot splits near the pilot burner face, displaying the mean temperature field with zero velocity line for natural gas and 40% H_2, and the mean equivalence ratio for 40% H_2. Pilot splits were normalised to the default value. Increased pilot split strengthens the CRZ and cools the wall, particularly at splits of 1.0 and 1.5, an effect unseen with natural gas. Higher pilot splits also lead to regions of high equivalence ratio near the wall, where unburnt fuel cools the burner face. Conversely, lower split yield near-stoichiometric conditions and higher temperatures. The differing diffusive properties of the H_2 blend likely contribute to fuel pooling near the burner face at higher pilot splits.

Figure 13 presents the effect of pilot split on combustion dynamic pressure amplitude at two distinct frequency ranges: 0 - 100 Hz and 600 - 1000 Hz. Natural gas and H_2 blends with 30% and 40% H_2 at full load were included in the analysis. Pilot splits were normalised relative to their default value, while pressure amplitudes were normalised against their design limits for each frequency range.

The reduction of pilot split had a negligible impact on combustion dynamic pressure amplitude for natural gas across both frequency ranges. For 30% H_2, decreasing the pilot split from its default setting led to a slight reduction in pressure amplitude within the 0-100 Hz range. However, a more significant reduction in pressure amplitude was observed for 40% H_2 at this frequency range. Increasing the pilot split from its default setting had minimal influence on pressure amplitude for both 30% and 40% H_2 blends. Within the 600-1000 Hz frequency range, pressure amplitude was relatively insensitive to pilot split variations for the H_2 blends as with natural gas. Notably, for 40% H_2 with a zero percent pilot split, pressure amplitudes remained low, reaching only 50% and 20% of their limits for the two frequency ranges. Operating with a zero percent pilot split for 40% H_2 offers the potential for safe operation with low NO_x emissions, low pilot burner surface temperature, and reduced combustion dynamic pressure.

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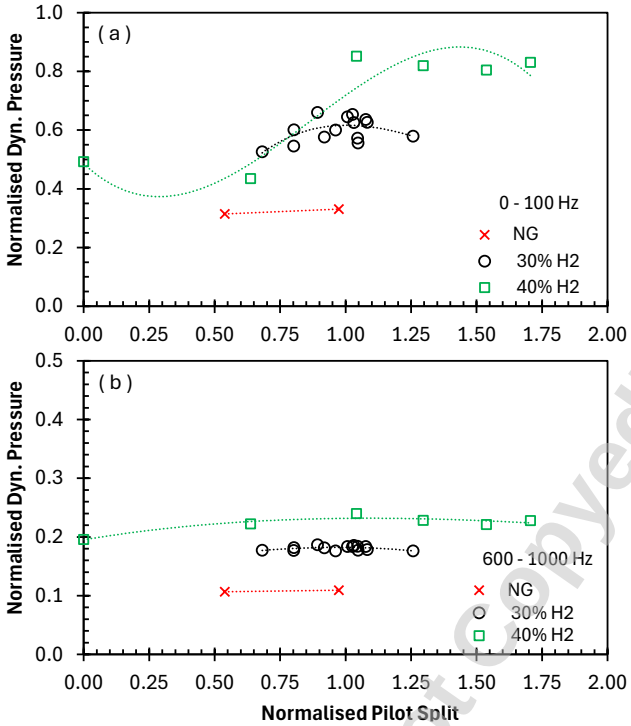

Figure 13 Effect of pilot split on combustion dynamic pressure amplitude (top: LFD [0 – 100 Hz], bottom: IFD [600 – 1000 Hz])

3.3 Effect of Ambient Temperature on Combustor Performance

The effect of ambient temperature on combustor performance was investigated for a 30% H_2 fuel blend across various load conditions under three different ambient temperatures: -20 °C, ISO, and +45 °C. The results are presented in Figure 14 for the impact on NO_x emissions and pilot burner surface temperature, and in Figure 15 for the impact on combustion dynamic pressure amplitude across two frequency ranges. NO_x emissions, pilot burner surface temperatures, and TET were normalised based on their values at ISO full load conditions, while pressure amplitudes were normalised against their design limits for each frequency range.

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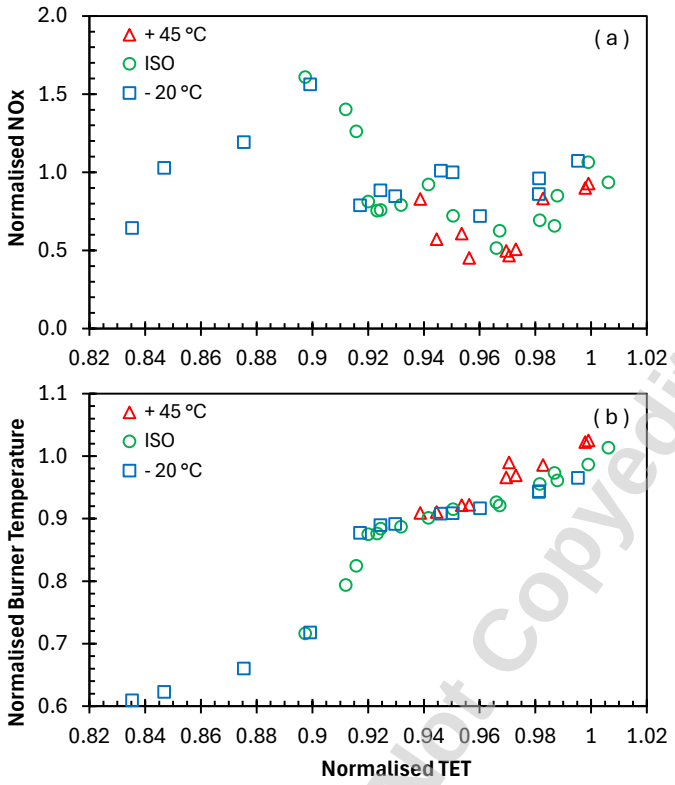

Figure 14 Ambient temperature impact on NO_x emission and pilot burner surface temperature for the fuel blend with 30% H_2

As illustrated in Figure 14(a), NO_x emissions were not significantly affected by ambient temperature, with marginally higher values observed at higher TETs for -20 °C ambient temperature compared to the other ambient temperatures. The ambient temperature had a negligible influence on pilot burner surface temperature across all TETs tested as shown in Figure 14(b).

As shown in Figure 15 (a), colder ambient temperatures led to higher combustion dynamic pressure amplitudes within the 0 - 100 Hz frequency range. However, these amplitudes remained within the design limit. For the 600 - 1000 Hz frequency range shown in Figure 15(b), ambient temperature had a negligible effect on combustion dynamic pressure amplitude, with only slightly higher amplitudes observed at -20 °C.

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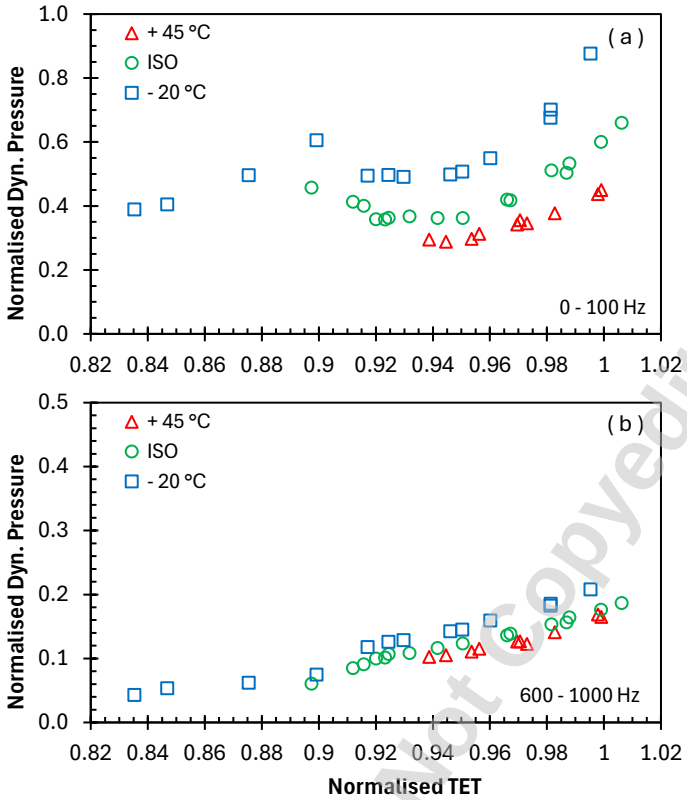

Figure 15 Ambient temperature effect on combustion dynamic pressure amplitude for the fuel blend with 30% H_2

3.4 Impact of H_2 Concentration on Turndown Control Temperature

As previously discussed, there is potential to optimise pilot split settings for 35% and 40% H_2 blends, aiming to lower the turndown control temperature while maintaining NO_x and CO emissions below 25 ppmvd and 10 ppmvd, respectively, at part load. To achieve this, pilot split settings were optimised for TET below 0.92 (normalised value), and test results under natural turndown operation are presented in Figure 16 for pilot split settings and Figure 17 for NO_x and CO emissions.

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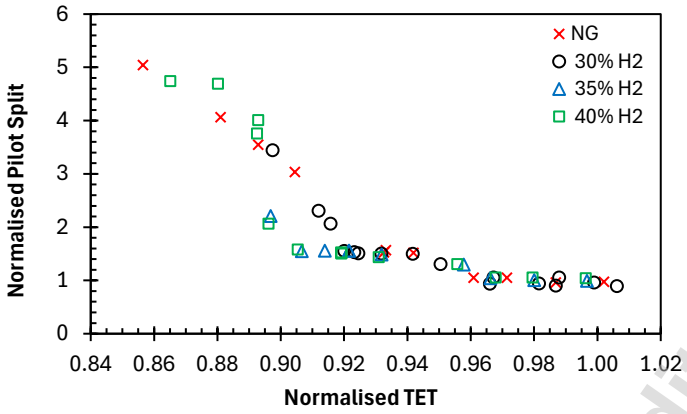

Figure 16 Optimised pilot split settings for 35% and 40% H_2 blends

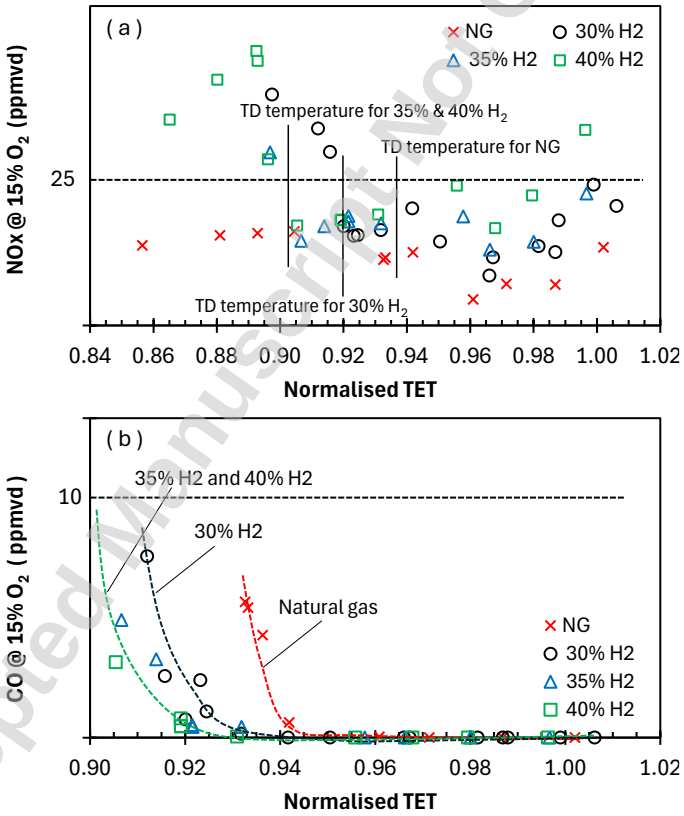

Figure 17 NO_X and CO emissions, and resulted turndown control temperatures

Compared to Figure 5, pilot splits for the TET range between 0.9 and 0.92 (normalised values) for 35% and 40% H_2 blends were reduced, as shown in Figure 16. As a result, and as shown in Figure 17,

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less than 25 ppmvd NO_x emissions were achieved, while CO emissions were unaffected (compare Figure 17 to Figure 6). This enabled a reduction in the turndown (TD) control temperature for these H_2 blends. Although there were no significant differences in pilot split setting and turndown control temperature between 35% and 40% H_2 blends, these values differed from those of 30% H_2 and natural gas. The results suggest that fuel blends with different H_2 concentrations may require distinct pilot split settings and turndown control temperatures to achieve lower emissions across a wider range of engine load conditions and improve thermal efficiency at part load.

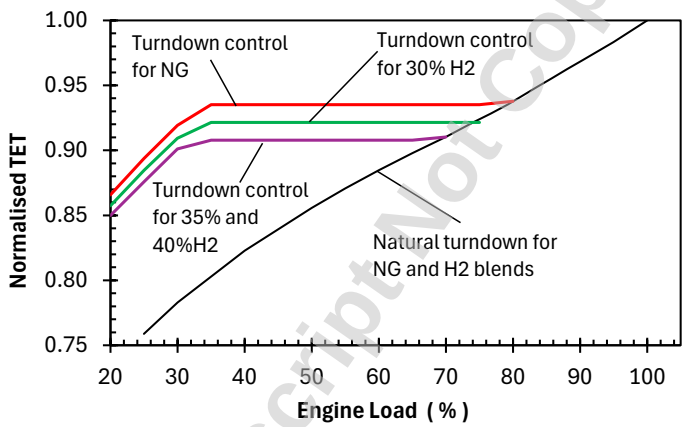

Figure 18 TET under natural turndown and turndown control operation across engine loads for natural gas and different H_2 blends

Figure 18 illustrates the TET (normalised values to its full load) across engine loads for natural turndown and with turndown control operation. For natural turndown operation, TET decreases as engine load decreases by lowering the fuel and air mass flow rates. At a certain load condition, known as the turndown point, CO emissions begin to increase exponentially due to TET approaching a critically low value. With turndown control, TET is kept constant as load is reduced by 'bleeding' the compressor air into the exhaust (or by variable guide vane for single shaft engines). By operating at a higher TET at part load when turndown control is enabled, CO emissions can be effectively controlled to low levels. Under these conditions, pilot split can also be kept low compared to the natural turndown at the same load range. Consequently, NO_x emissions can be effectively reduced.

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As shown in Figure 18, compared to fuel blends with H_2, natural gas has a higher turndown control temperature. The turndown control temperature is lowered with increasing H_2 content in the fuel blend. By operating at a lower turndown control temperature, part load engine thermal efficiency is enhanced.

NO_x and CO emissions for simulated engine operation with turndown control temperatures shown in Figure 18 are presented in Figure 19 for natural gas and 30% H_2 fuel blend. Compared to natural turndown operation, NO_x and CO emissions could be controlled below 25 ppmvd and 10 ppmvd, respectively, for a wider engine load range.

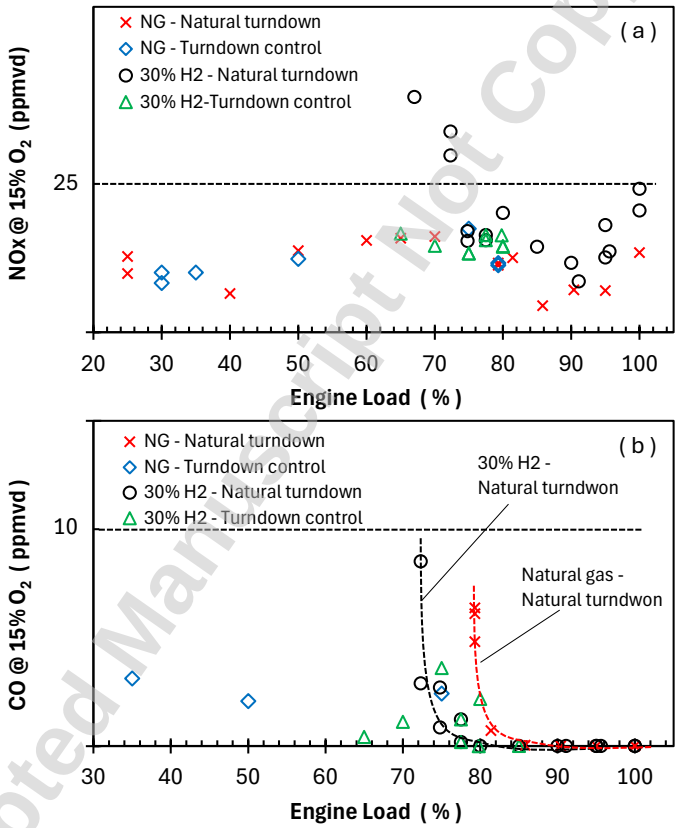

Figure 19 NO_X and CO emissions across engine loads under natural turndown and turndown control operation for natural gas and 30% H_2 blend

3.5 Overfiring Tests

To evaluate the robustness of the combustion system when operating with H_2 containing fuels, particularly during transient engine operation, the combustor was overfired by increasing fuel flow at

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75%, 85%, and 100% load operating conditions. This testing was conducted with a fuel blend containing 30% H_2. Additionally, fuel blends with 30% and 35% H_2 were tested at an elevated load condition of 105% to assess operating margins.

Table 1 Overfiring operating conditions

Test Point	Operating Condition
1	100% engine load with 30% H_2
2	10 K TET overfiring from point 1
3	33 K TET overfiring from point 1
4	105% engine load with 30% H_2
5	105% engine load with 35% H_2

Figure 20 presents the results for the five test points listed in Table 1 for overfiring operation. Point 1 represents 100% engine load with 30% H_2 as a reference. Point 2 indicates overfiring by increasing fuel flow to achieve a 10 K increase in TET from point 1. Point 3 represents overfiring with a 33 K TET increase from point 1. Point 4 corresponds to enhanced engine load at 105% with 30% H_2. Finally, point 5 represents the 105% enhanced engine load with 35% H_2 to evaluate operating margins. Presented NO_x emissions and pilot burner surface temperatures were normalised at those values measured at point 1.

Both NO_x emissions and pilot burner surface temperatures increased when the combustor was overfired, but burner temperatures remained within the design limit. There is a substantial increase in NO_x emissions for the 10K and 33K overfired TET conditions due to significantly higher flame temperature. At elevated engine load conditions of 105%, NO_x emissions and pilot burner surface temperatures were higher compared to 100% load conditions with a 30% H_2 fuel blend. These values increased further with a 35% H_2 fuel blend, but the pilot burner surface temperature remained within

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the design limit. The results demonstrate that the combustor design is robust and can withstand overfiring and elevated loads.

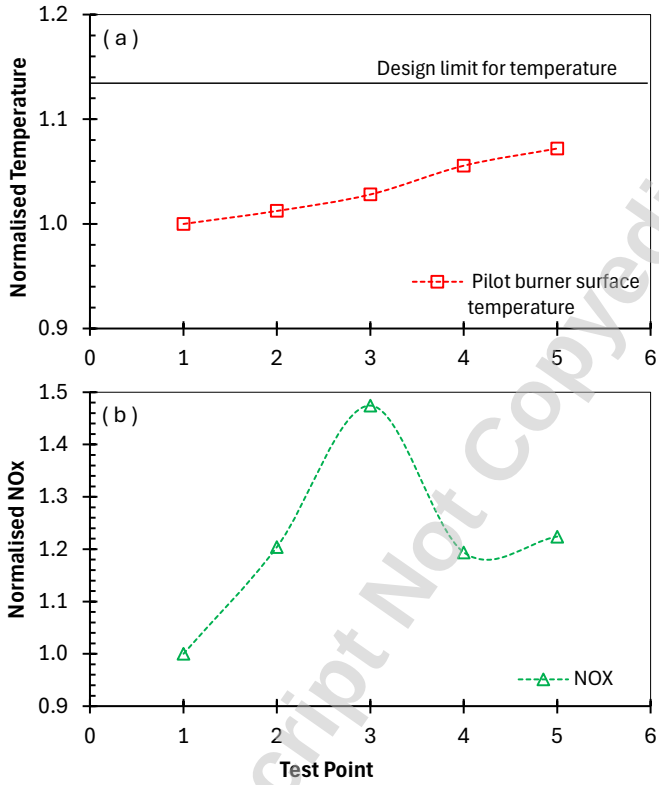

Figure 20 Pilot burner surface temperature and NO_x emissions at overfired conditions

4. CONCLUSIONS

The impact of H_2 concentration on the performance of the Siemens Energy SGT-300-2S 9 MW industrial gas turbine combustor was experimentally investigated through a high-pressure combustor rig test at engine operating conditions

H_2 concentrations up to 40% were tested under natural turndown operation. NO_x emissions were increased and the CO kick-off point was delayed as H_2 concentration in the fuel blend was increased. The surface temperature of the pilot burner and the amplitudes of combustion dynamic pressure at two distinct frequency ranges increased with increasing H_2 concentration, but this increase was only noticeable at higher engine load conditions. The combustor can wall temperatures at most locations

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were found to be insensitive to H_2 concentration. An increase in the wall temperature at the prechamber was noticeable at the 40% H_2 enrichment condition due to an upstream shift in the flame position

The influence of pilot split on combustor performance at full load was also investigated. As expected, reducing pilot split was found to lower NO_x emissions and pilot burner surface temperature for natural gas. A similar trend was observed for NO_x emissions for the H_2 blended fuels. However, lowering pilot split was found to increase the pilot burner surface temperature. Ambient temperature was observed to have limited impact on emissions and pilot burner surface temperature but increased the combustion dynamic pressure amplitude at colder ambient temperature.

Finally, the effect of H_2 concentration on turndown control temperature was demonstrated. With higher H_2 concentrations, it was found that the turndown control temperature could be lowered, leading to enhanced part load engine thermal efficiency. The combustion system's robustness was tested under overfired and elevated load conditions demonstrating stable combustion and proving its reliability.

A key observation in this study is that the optimal pilot split and turndown control temperature can vary significantly depending on the specific fuel composition. This finding underscores the importance of optimising these parameters for each fuel blend to ensure stable and efficient combustion operation.

The test results showed that the SGT-300-2S combustion system can safely operate with up to 40% H_2 in the fuel blend without experiencing flashback or hardware damage. When using 30% H_2, NO_x emissions of 25 ppmvd and CO emissions of 10 ppmvd can be achieved under turndown control operation across a wide range of engine load conditions. The effect of H_2 enrichment on combustor exit temperature distribution is numerically studied and discussed in part II paper and will be a future validation study on the high-pressure combustor test rig.

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NOMENCLATURE

B2E Bleed to Exhaust

CEC Clean Energy Center

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CRZ	Central Recirculation Zone	
DLE	Dry Low Emissions	
Dyn.	Dynamic	
GT	Gas Turbine	
IFD	Intermediate Frequency Dynamics	
ISO	Standard condition at 1.0135 bar and 15 °C	
LBO	Lean Blow Out	
LFD	Low Frequency Dynamics	
LHV	Lower Heating Value	
NG	Natural gas	
ORZ	Outer Recirculation Zone	
pmvd	parts per million on a volumetric and dry basis	
RMS	Root Mean Square	
SEITL	Siemens Energy Industrial Turbomachinery Ltd.	
TD	Turndown	
TET	Turbine Entry Temperature	
UHC	Unburnt Hydrocarbon	
VGV	Variable Guide Vane	
WI	Wobbe Index	MJ/m³

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